

ASCENT-3 — Statistical Analysis Plan (Synopsis)

Protocol RZB441-UC-301 | Phase III, randomized, double-blind, placebo-controlled trial of RZB-441 (roزالibumab) in moderate-to-severe ulcerative colitis | SAP Version 3.0 | Status: FINAL, approved pre-database-lock

This SAP synopsis is the controlling methodological reference for the Week-24 interim integrity re-analysis. Where any working draft, summary table, or export disagrees with the methods defined here, this SAP and the subject-level source data govern.

1. Study Design Parameters

Parameter	Specification
Design	Randomized 1:1, double-blind, placebo-controlled, parallel-group
Treatment arms	RZB-441 (active) and Placebo — these are the ONLY two valid arm codes
Randomization stratification factors	Geographic region (3 levels) and prior biologic exposure (Naive / Experienced)
Planned randomized N	480 (240 per arm)
Study start (first subject randomized)	2024-01-15
Interim data cutoff (DSMB)	2025-09-30 — records randomized after this date are not part of the interim
Primary endpoint	Time to confirmed clinical remission by Week 24 (Day 168)
Key secondary endpoint	Clinical response at Week 8 (binary)
Pre-specified MCID (remission rate diff.)	10.0 percentage points

2. Analysis Populations and Record-Validity Rules

Intention-to-treat (ITT) — PRIMARY analysis population. All validly randomized subjects, analyzed as randomized. ITT is the primary population for all efficacy endpoints. The Per-Protocol (PP) set is supportive only and must NOT be substituted for ITT in the primary efficacy or go/no-go assessment (ICH E9(R1)).

Record-validity rules (applied before any population is defined). A record is invalid and must be excluded from ALL analysis populations if any of the following holds. The data-management query log (ascent3_data_query_log.csv) documents each affected subject:

#	Invalidity rule	Evidence source
1	Duplicate subject_id (retain first valid occurrence only)	subject_level
2	randomization_date after the 2025-09-30 interim cutoff	subject_level
3	arm not equal to 'RZB-441' or 'Placebo' (incl. blank / 'Screen Failure' / off-protocol dose)	subject_level
4	death_date earlier than randomization_date	subject_level
5	subject_id absent from the IWRS randomization schedule	randomization_schedule
6	any visit_date earlier than randomization_date	visit_records

Per-Protocol (PP) set. ITT minus subjects with a major protocol deviation flagged 'PP-EXCLUDE' in the query log (prohibited concomitant biologic, study-drug compliance < 80%, randomized but never dosed, or wrong stratum entered). Major deviations remain in ITT.

3. Primary Endpoint — Estimand and Estimation Method

Estimand. Cumulative incidence of confirmed clinical remission by Week 24 (Day 168) in the ITT population, by arm, under a treatment policy strategy (ICH E9(R1) §2), with the treatment effect expressed as the between-arm remission-rate difference (RZB-441 minus Placebo).

Estimation. The primary endpoint is a binary outcome (confirmed remission yes/no) assessed at a fixed time window ending at Day 168. Because enrollment is staggered over time and subjects may discontinue study treatment or be lost to follow-up before Day 168, the estimation method must be selected to be appropriate for the data structure and consistent with the estimand framework in ICH E9(R1). The selected method must account for subjects with incomplete follow-up. Report each arm's Week-24 remission rate with a two-sided 95% confidence interval and the between-arm rate difference with its 95% CI. See External Links for methodological references on estimation of proportions in the presence of incomplete follow-up.

4. Primary Success (Go / No-Go) Decision Rule

The interim outcome is declared **GO** only if BOTH conditions are met, evaluated on the primary Week-24 remission-rate difference derived by the SAP-compliant estimation method:

Condition	Requirement
(i) Statistical superiority	Lower bound of the two-sided 95% CI for the primary remission-rate difference is > 0
(ii) Clinical relevance	Point estimate of the primary remission-rate difference is ≥ 10.0 percentage points (MCID)

If condition (i) holds but condition (ii) does not, the result is a statistically significant but not clinically meaningful effect and the decision is **NO-GO**.

5. Secondary Endpoint — Week-8 Clinical Response

Week-8 clinical response (binary) is analyzed in the ITT population. Baseline disease severity (Moderate vs Severe, from baseline total Mayo score) was **NOT** a randomization stratification factor. As a strong prognostic factor that was not balanced by design, it may differ materially between arms and confound the observed Week-8 response rates. A pre-specified analysis that appropriately accounts for the baseline severity distribution is required. Report both the crude pooled rate difference and the severity-adjusted estimate; where the two diverge materially, identify the reason and document which estimate governs the interpretation.

6. Multi-Center Pooling

When combining results across sites, the pooling approach must be appropriate given the variability in site sample sizes. An analyst preparing a site-level summary should document the method used to derive the overall rate and confirm it is consistent with standard practice for multi-center clinical trials.

7. Sample-Size Methodology for the Confirmatory Study

The planned 52-week confirmatory study is sized for the two-proportion (remission rate) comparison using:

$$n \text{ per arm} = (Z_{1-\alpha/2} + Z_{1-\beta})^2 \times [p_1(1-p_1) + p_2(1-p_2)] \div (p_1 - p_2)^2$$

Assumption	Value
Two-sided alpha	0.05 (Z = 1.9600)
Power	0.9 (Z = 1.2816)
Dropout inflation	20% (divide computed n by 0.80)
Originally assumed remission rates	p1 = 0.40 (RZB-441), p2 = 0.25 (Placebo)
Re-estimation requirement	Re-size using the corrected Week-24 remission rates from this interim re-analysis

Round each computed per-arm n UP to the next whole subject before and after dropout inflation.

8. Regulatory References

ICH E9(R1) Addendum on Estimands and Sensitivity Analysis (analysis populations, ITT principle, estimands). CONSORT 2010 Statement (subject disposition / flow accounting and reporting of exclusions and baseline imbalance). See the task External Links for URLs.